DOES HYPERBOLIC GEOMETRY HELP GRAPH-BASED SENTENCE POOLING? A CONTROLLED STAGE-WISE STUDY

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ABSTRACT

Hyperbolic space represents tree-like structure with less distortion than flat space, and transformer token states are often described as tree-like. That pairing has motivated hyperbolic versions of many models. We ask whether it actually helps, on one architecture, under controlled conditions.

Our subject is GLOT (Mantri et al., 2026), a sentence pooler that builds a graph over the tokens of a frozen language model, refines it with a GNN, and aggregates it into a single vector. Geometry enters in exactly three places — how the graph is built, how messages combine, how states are pooled — so we replace each independently and give all eight combinations the same search budget over fifteen shared seeds.

The three parts disagree, and that is the main result. The hyperbolic readout is reliably harmful, on an encoder and again on a decoder, and we can say why: it amplifies seed variance rather than acting as a useful prior. Hyperbolic message passing is the only part that ever helps. The hyperbolic graph does nothing anywhere. Reported as a single "hyperbolic" switch these would have averaged to about zero; the stage-wise design is what makes them visible.

We are careful about the one gain. It is significant end to end, but a factorial that separates curvature from the graph density and architecture it was tuned alongside cannot attribute it to either — the total is significant and none of its components is. We report it as real but unexplained.

We also question the premise. The usual justification for going hyperbolic is a low measured δ-hyperbolicity. That statistic is an artifact of a few very large token norms: on an outlier-free metric the backbones it is used to tell apart are indistinguishable (0.2127 against 0.2114), while the Euclidean version differs between them by two orders of magnitude. Measured tree-likeness cannot tell you where curvature will help — only a controlled, component-wise comparison can.

1 INTRODUCTION

Turning a language model's token states into one vector is a first step for most sentence tasks. Simple poolers — mean, max, `[CLS]` — treat tokens as an unordered bag. A newer idea is to treat pooling as *relations first, aggregation second*: build a graph over tokens, refine it with a GNN, then aggregate. GLOT (Mantri et al., 2026) is a strong example, matching competitive GLUE and MTEB numbers with $20\times$ fewer trained parameters and a frozen backbone. Our runs support the claim it rests on: on the authors' own noise test, GLOT beats every bag-of-tokens pooler by 14–35 points in every seed.

This paper asks what else that design can carry. Hyperbolic space is a natural candidate for all three of its parts, so we build a hyperbolic version of each and measure them separately. That separation is the point: two of the three move in opposite directions, so a single "hyperbolic" switch would have reported their average and found nothing.

Contributions.

1. **A stage-wise hyperbolic study** (§4). Under an equal search budget with 15 shared seeds, the hyperbolic readout (B) is significantly harmful on an encoder and on a decoder; hyperbolic message passing (C) is the only part that ever helps; the hyperbolic graph (A) is flat everywhere. We report what each part does, not what they do on average.
2. **The usual motivation does not survive measurement** (§4). Low δ-hyperbolicity is driven by a handful of very large token norms. On an outlier-free metric the backbones it is used to tell apart are indistinguishable. So measured tree-likeness cannot tell you where curvature will help.
3. **The choices that changed our answers** (§3). A cache-warming order worth 5 MCC, a search that reverses arm rankings, a threshold that silently produces a complete graph on RoBERTa, and two defects in the released code. We list these because each of them cost us an experiment.

2 What we changed

Let $x_1, \dots, x_L$ be the token states of a frozen backbone. GLOT joins two tokens when their cosine passes a threshold,

$$A_{ij} = \mathbf{1}[\,\cos(x_i, x_j) > \tau\,], \tag{1}$$

runs K graph-attention layers over the result, and aggregates with a learned readout. Only the pooler trains. Geometry enters in three places — how the graph is built, how messages combine, how states are aggregated — so we replace each in turn. Appendix A gives the Poincaré ball, the maps $\exp_0^c$ and $\log_0^c$, and the geodesic distance we use. Figure 1 in Appendix B shows where each stage sits.

Stage A: a Poincaré graph. Lift the tokens into the ball and join them when their geodesic distance falls below the q-quantile of all pairwise distances, $A_{ij} = \mathbf{1}[\,d_c(\exp_0^c(\tilde{x}_i), \exp_0^c(\tilde{x}_j)) < \rho_q\,]$. Using a quantile rather than a fixed radius fixes the edge count at q, so A and the cosine baseline can be compared at the same sparsity.

Stage B: an Einstein gyro-midpoint readout. Replace the weighted sum with the Einstein midpoint in Klein coordinates,

$$m_K = \frac{\sum_i a_i \gamma_{k_i} k_i}{\sum_i a_i \gamma_{k_i}}, \qquad k = \frac{2x}{1 + c\|x\|^2}, \qquad \gamma_k = (1 - c\|k\|^2)^{-1/2}, \tag{2}$$

then map back and into the tangent space for the task head. Unlike a plain mean this depends on curvature: as $c \to 0$ it becomes the arithmetic mean.

Stage C: Möbius message passing. Replace each GNN layer with a hyperbolic one that lifts to the tangent space to combine neighbours,

$$h_i' = \exp_0^c\Big(\sigma\Big(\textstyle\sum_{j \in \mathcal{N}(i)} \alpha_{ij}\ \log_0^c\left(W \otimes_c h_j\right)\Big)\Big), \tag{3}$$

with $W \otimes_c h = \exp_0^c(W \log_0^c(h))$ and α_{ij} either degree-based (HGCN) or attention (HGAT).

Curvature is searched in a scale-free way. Curvature and feature scale cannot be told apart here. The geodesic distance depends only on $\sqrt{c}\,\|x\|$, so doubling the features and quartering c leaves every distance unchanged, and searching c on its own wastes most of the grid. We instead rescale features to unit mean norm and search the *effective radius* $r_{\text{eff}} = \sqrt{c}\,\mathbb{E}\|x\|$ over $\{0.5, 1, 1.5, 2, 3\}$. This is not cosmetic: with raw BERT token norms (≈ 14.7) the map $\exp_0^c$ saturates to radius 1.0 in `float32`, real pairwise distances land in $[8.85, 11.76]$, and any fixed distance threshold below about 9 returns an empty graph (Appendix M).

We run all eight combinations of A, B and C.

3 How we ran it, and the choices that changed the answer

Every arm is frozen-backbone, 2 epochs, Adam, batch 32; only the pooler trains. Tuning draws 40 configurations at one held-out seed. Confirmation re-runs the winner on the fixed seed set

$\{1, \ldots, 15\}$, shared across arms, so every comparison is paired. We report the exact two-sided sign test beside the paired t-test and call something significant only when both agree, with a Bonferroni correction across arms ($\alpha = 0.0063$). Full protocol in Appendix C. Five choices moved our results more than the effects we were trying to measure.

The search space must be the same for every arm. An earlier campaign of ours searched only graph knobs and left the learning rate, depth and width pinned at values picked for a *flat* pooler. That handed the baseline an advantage and produced a false positive we retract in Appendix K: Stage A on STS-B looked like $+0.223$ Spearman with $15/15$ seeds and $t(14) = 6.20$, and collapsed to $+0.028$ once the baseline was allowed to pick $K = 4$. Everything reported here uses GLOT's full grid applied identically to all arms: 40 trials each, 15 seeds, 990 runs.

The search really does use that freedom. The learning rate does not settle on one value. On STS-B all nine arms chose 2×10^{-4}; on CoLA eight of nine chose 1×10^{-3}; MRPC and RTE split. Depth, width and threshold move too. Arms were therefore not locked to a rate chosen for a flat pooler, which is the main form of the objection that non-Euclidean layers get judged under flat-tuned optimisers. Appendix D lists the selected configurations.

Tuning maxima are inflated, and can reverse a ranking. With T trials and per-seed noise σ, the best of T sits above the truth even for an arm identical to the baseline — about 1.83 here on STS-B. On the RoBERTa campaign the inflation is large enough to flip the order outright: the quantile correction is first of five at the tuning maximum (56.77) and last of five over 15 seeds (53.19). Reporting tuning maxima would have inverted our conclusion. We report confirmation means only.

Cache order is worth about 5 MCC. The released code caches backbone states and builds the shuffled loader *before* the classifier is created. On a cache miss the loader consumes the global RNG; on a hit it does not. The same seed therefore gives a different initialisation depending on cache state — 40.37 cold against 45.54 warm on CoLA, about six times the seed standard deviation. We build all caches before any arm runs. Any reproduction that skips this will shift by several points.

The estimator matters as much as the method. Mantri et al. (2026) fix seed 42 and report the best τ per task on the split they also report, so every published number is one draw maximised over a search. We report 15-seed means. On CoLA the choice of estimator alone moves the same runs by nearly 5 MCC (Appendix E). Matching their estimator exactly, we reach 50.52 MCC on CoLA against a published 47.49 and 59.57 on RTE against 59.21; STS-B stays 0.87 short and we could not explain it. Every comparison in this paper is against our own baseline under identical conditions, never against a published number.

4 What we found

Appendix F holds the numbers; this section states what they say.

The hyperbolic readout is harmful, and this is the firmest result in the paper. Every arm containing Stage B loses 0.75–1.24 Spearman on STS-B, with 0 or 1 of 15 seeds positive. B alone gives $t = -17.00$ with $0/15$ seeds and a sign-test p of 6×10^{-5}, the smallest value 15 paired seeds can produce. All four arms containing B survive Bonferroni. The result repeats on a decoder backbone, TinyLlama, where AB costs 3.56 Spearman and ABC costs 4.51, both $0/15$.

There is a mechanism, not just a mean. On the decoder the seed standard deviation rises in step with the number of hyperbolic parts: $0.48 \to 0.95 \to 1.87 \to 3.58$, a $7.5\times$ spread. The gyromidpoint behaves like a variance amplifier rather than a useful prior. It is also the one component whose theoretical motivation is weakest here: pooling averages tokens that the graph has already smoothed, and the Einstein midpoint's curvature dependence pulls hardest exactly on the large-norm tokens that Appendix G shows are outliers.

One task disagrees, and we state it rather than bury it. On RTE all four B-containing arms are *positive* ($+0.79$ to $+1.85$ accuracy). None is significant, and none could be: RTE has 277 evaluation examples, a seed standard deviation of 3.36, and a minimum detectable effect of 2.7 points —

larger than any effect reported in this paper. We read RTE as uninformative rather than as contrary evidence, but it does mean the honest claim is "B is harmful on the three tasks that can measure it", not "B is harmful everywhere".

Message passing is the only part that ever helps. Stage C reaches $+1.42$ MCC on CoLA with $13/15$ seeds positive ($t = 4.11$, sign $p = 0.0074$) — significant on both tests, though just above our Bonferroni line of 0.0063. On STS-B, A+C is the best of the nine arms ($+0.24$) and C is second ($+0.20$). Every arm with the hyperbolic GNN and *without* the readout is positive on both tasks.

We do not present this as settled. It does not reach MRPC, where no arm beats the baseline and every hyperbolic arm is slightly negative (-0.15 to -0.33, none significant). It does not separate from the graph-free control on CoLA ($+0.58$, $8/7$, $p = 1.00$). And, most importantly, it does not decompose.

The search had given C a $3.6\times$ denser graph than the baseline ($q = 0.38$ against 0.10) together with half the depth and half the width, so the $+1.42$ mixed geometry with those choices. We ran the 2×2 that separates them on the same 15 seeds (Table 7). Hyperbolic message passing on the baseline's *own* graph and architecture gains $+0.40$ ($8/7$, $p = 1.00$). The denser graph and smaller network with *Euclidean* message passing gains -0.01 ($10/5$, $p = 0.30$). The remaining $+1.03$ is an interaction — and it too is not significant ($t = 1.35$, $10/5$, $p = 0.30$).

So the decomposition is exact and useless: the three components sum to $+1.416$, the total is significant, and no part of it is. The component contrasts carry 2–3$\times$ the variance of the total (sd 1.75, 2.55, 2.95 against 1.33), because each differences two independently trained cells. Resolving the interaction at 80% power would need roughly 65 seeds; we ran 15. We therefore report Stage C's gain as real but *unattributed*: it needs both the denser graph and the curved message passing, and we cannot say which does the work. That C is negative on MRPC, where the search chose a $15\times$ *sparser* graph, is consistent with needing both, but is not evidence for it.

On these tasks the edges are doing less than expected. The graph-free arm is not worse than the baseline on any of the four tasks (CoLA $4/11$, STS-B $8/7$, MRPC $6/9$, RTE $7/6$), and on CoLA it ranks second of eight. It is still a trained pooler, so this is not a claim that GLOT's design is unnecessary — against real bag-of-tokens poolers the graph wins by 14–35 points on the authors' noise test. The narrower claim is that on GLUE, at 40 trials per arm, accuracy does not come from the edges. That matters here because it sets the bar every geometric claim must clear: Stage C clears it on STS-B and not on CoLA, and Stage B is significantly *below* it.

The graph is flat, and so is the choice of GNN. Stage A moves nothing anywhere: $+0.70$ on CoLA, $+0.03$ on STS-B, $+0.04$ on MRPC, $+0.04$ on the decoder, none significant. Swapping GAT for GCN or GIN moves CoLA by at most 0.52 MCC and STS-B by 0.23, inside seed noise. Neither the geometry of the graph nor the architecture that reads it is where the variation lives.

The components cancel. B and C carry opposite signs. A study that turned "hyperbolic" on as one switch would have reported their average, which is close to zero, and concluded that geometry does not matter here. It does — the parts simply disagree, and only a stage-wise test can see that.

The motivation is an artifact. Hyperbolic layers are usually justified by measuring Gromov δ-hyperbolicity and finding it small. On the Euclidean metric ModernBERT at layer 16 gives $\delta = 0.0021$, which reads as almost perfectly tree-like. But that layer has a maximum-to-median token norm ratio of 156, and δ is scale-sensitive: a few very large "attention sink" tokens dominate it and pull it toward zero. On the angular metric $\arccos(\cos(\cdot, \cdot))$, which ignores those norms, ModernBERT gives 0.2127 and BERT gives 0.2114 — indistinguishable, while the Euclidean numbers differ by two orders of magnitude. Flan-T5 shows the same pattern with $357\times$ norm outliers. So measured tree-likeness cannot tell you which backbone or which component will benefit from curvature. We did not use it to choose; we applied all three stages everywhere and let the paired tests decide.

Porting the method to a new backbone. Equation 1 compares a cosine to a fixed number, which only makes sense if cosine scale is comparable across models. It is not. On RoBERTa the *tenth*-percentile token cosine is 0.693, above the largest τ in the published grid, so every searched setting yields an essentially complete graph (0.991 density). The useful finding is what happens next: GLOT is unharmed by that dense graph, while the obvious repair — switching to a quantile threshold — costs 1.41 MCC and survives Bonferroni. It also won its tuning stage before finishing last over 15 seeds, so selection actively favoured it. We therefore withdraw density-matching as a recommendation for anyone porting this method.

Scope, and one confound it creates. Mantri et al. (2026) report ten GLUE tasks. We report four (CoLA, STS-B, MRPC, RTE), plus RoBERTa, ModernBERT, SmolLM2 and TinyLlama in supporting roles. The four we chose are the four *smallest*, and not by accident: our design costs 495 runs per task, which is 22 hours for CoLA but an estimated 38 and 41 days for QQP and MNLI, with 149 GB of cached states between them.

That leaves a confound we cannot currently rule out. Every task here has at most 8,551 training examples, and Stage B's failure mode is variance amplification — a quantity that shrinks as data grows. Our Stage B negative may therefore be a small-data effect rather than a property of the geometry. SST-2 (67k examples, $7.9\times$ CoLA) is the cheapest test that would settle it, and we did not run it. We do report IMDB and seven MTEB tasks (Appendix L), but only at one seed and only for the graph stage, so they carry no weight here.

5 Conclusion

Whether hyperbolic geometry helps graph-based pooling depends entirely on *which part* receives it, and the parts disagree enough to cancel.

The Einstein gyro-midpoint readout is reliably harmful: negative in every setting we ran, 0 or 1 of 15 seeds positive on STS-B, surviving Bonferroni on an encoder and repeating on a decoder, with a variance signature that suggests why. Möbius message passing is the one part that ever helps, by $+1.42$ MCC on CoLA, but it does not reach MRPC and the comparison is not yet clean. The Poincaré graph does nothing at all.

Two lessons travel beyond this architecture. The standard reason to go hyperbolic — a low measured δ — is an artifact of norm outliers, and on an outlier-free metric the backbones it distinguishes are the same. And a comparison between geometries is only as good as the budget given to its control: our own retracted result, and the confound we found in our headline number, both come from arms that were not given the same freedom. We report both so the next study does not repeat them.

Reproducibility statement

All confirmation results use the fixed seed set $\{1, \ldots, 15\}$, shared across arms so comparisons are paired; tuning uses one held-out seed. Every run appends a row to a per-campaign CSV recording the configuration, the realised edge density, the per-epoch metrics and the wall-clock time. These logs are released unmodified, including runs that crashed or produced empty graphs, and the analysis scripts behind every table read those CSVs directly. Caches are built before any arm runs; that ordering is load-bearing, not an optimisation (§3).

References

Krishna Sri Ipsit Mantri, Carola-Bibiane Schönlieb, Zorah Lähner, and Moshe Eliasof. Towards improved sentence representations using token graphs. In *International Conference on Learning Representations (ICLR)*, 2026. Verified against extracted PDF text; add URL/pages before submission.

A Hyperbolic preliminaries

We work in the Poincaré ball $\mathbb{B}^d_c = \{x \in \mathbb{R}^d : c\|x\|^2 < 1\}$ of curvature $-c$, with Möbius addition $\oplus_c$ and the maps

$$\exp_0^c(v) = \tanh\big(\sqrt{c}\,\|v\|\big)\frac{v}{\sqrt{c}\,\|v\|}, \qquad \log_0^c(x) = \operatorname{artanh}\big(\sqrt{c}\,\|x\|\big)\frac{x}{\sqrt{c}\,\|x\|}, \tag{4}$$

and geodesic distance $d_c(x, y) = \frac{2}{\sqrt{c}} \operatorname{artanh}\left(\sqrt{c}\,\|-x \oplus_c y\|\right)$. With features rescaled to unit mean norm, $c = r_{\text{eff}}^2$.

B Architecture

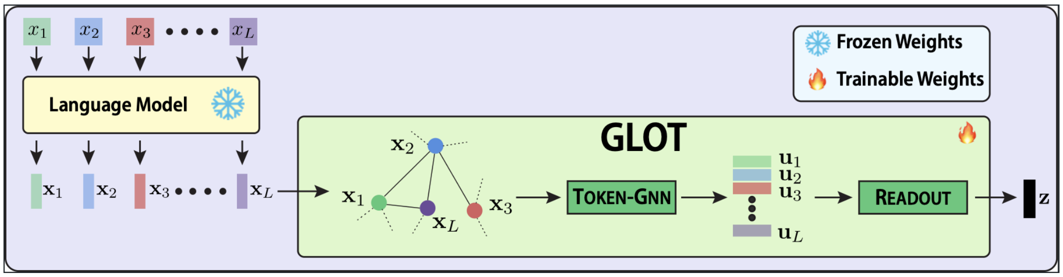

Figure 1: GLOT's pooling pipeline, reproduced from Mantri et al. (2026). Token states from a frozen backbone are thresholded into a graph (Eq. 1), refined by a token-level GNN, and aggregated into a single sentence vector $\mathbf{z}$; only the shaded block trains. Each of our three changes replaces exactly one depicted component: Stage A the graph, Stage B the readout, Stage C the token-GNN.

C Protocol

Frozen backbone, 2 epochs, Adam, weight decay searched, batch size 32; the pooler is the only trained component. Confirmation uses seeds $\{1, \ldots, 15\}$ shared across arms; tuning uses one disjoint held-out seed.

Matched budget. Every arm gets 40 trials in the wide campaign (10 in the earlier graph-only ones). The cosine baseline has a smaller native space, so we widen its grid to keep selection pressure equal. Equal trials is not equal *coverage*: 40 trials cover 6.2% of a 648-point space but under 0.01% of ABC's 4.98×10^7-point space. This asymmetry works against the complex arms, and we note it as a caveat against our own negative results rather than in support of them.

Testing. We report the exact two-sided sign test beside the paired t-test because evaluation-set quantisation makes the parametric standard error look too small (MRPC accuracy moves in steps of 0.245 on 408 examples). Both tests must agree. Bonferroni over the 8 arms tested against a common baseline gives $\alpha = 0.0063$. The minimum detectable effect at $n = 15$ is 0.225 (STS-B), 1.153 (CoLA), 0.317 (MRPC), 2.7 (RTE) and 0.503 (TinyLlama). RTE's figure exceeds every effect in this paper, so that task can only ever return a null; we report it, and draw nothing from it.

D Selected hyper-parameters

Each arm tunes its own configuration; these are the winners then run on all 15 confirmation seeds. Two things are worth reading off this table. The learning rate does not converge to a single value across tasks, so no arm was pinned to a rate chosen for a flat pooler. And on CoLA, Stage C selected a much denser graph ($q = 0.38$) and a smaller network than the baseline ($q = 0.10$, twice the depth, twice the width) — the confound discussed in §4. On MRPC the same search went the other way, giving C a $15\times$ *sparser* graph than the baseline; C is negative there. C wins where it picked dense and loses where it picked sparse, which is why the matched comparison is necessary.

Other selected settings vary similarly. Jumping-knowledge mode is uniform *within* a task but differs *between* them — `max` on CoLA, `cat` on STS-B — so it tracks the task, not the arm. Curvature, where it applies, spans the whole searched range $\{0.25, 1.0, 4.0\}$ with no consistent preference.

Table 1: Confirmed configurations, BERT wide campaigns. "q" is `tau_quantile` for cosine arms and `rho_quantile` for Poincaré arms; both mean "keep the closest q fraction of token pairs". "width" is the GNN hidden dimension, "proj" the projection dimension.

Task	Arm	lr	layers	width	proj	q
CoLA	baseline	1×10^{-3}	4	256	128	0.10
	A	1×10^{-3}	4	256	256	0.05
	B	2×10^{-4}	2	256	512	0.38
	C	1×10^{-3}	2	128	256	**0.38**
	AC	1×10^{-3}	2	128	256	0.05
STS-B	baseline	2×10^{-4}	4	64	256	0.20
	A	2×10^{-4}	4	64	256	0.10
	B	2×10^{-4}	4	64	128	0.38
	C	2×10^{-4}	2	128	128	0.20
	AC	2×10^{-4}	4	128	512	0.20
MRPC	baseline	2×10^{-4}	2	128	128	0.38
	B	1×10^{-3}	4	128	256	0.10
	C	2×10^{-4}	2	128	256	**0.025**

Table 2: One method, four estimators, on CoLA. Each row is the *same* pooler; only the way the score is computed changes. A dash means the knob does not apply. The published value sits between our mean and our maximum, and the row that matches how it was computed exceeds it.

Seeds	Aggregated	Selected over	CoLA MCC
Published (Mantri et al., 2026)			
1 (seed 42)	—	best τ of 3	47.49
Ours, same method			
15	mean	—	45.37
15	max	—	48.17
1 (seed 42)	—	best of 40 configs	**50.16**

E REPRODUCTION

The published τ-sensitivity curve cannot be read as a τ effect: it spans 7.87 MCC on CoLA, while we measure a spread of 5.46 across seeds at *fixed* τ.

F MAIN RESULTS

Table 4 gives the absolute scores behind every delta in this paper. It is laid out like Table 1 of Mantri et al. (2026) — rows are methods, columns are tasks, blocks are backbones — but the rows are our arms rather than competing poolers, because our question is which *component* of GLOT does what, not whether GLOT beats mean-pooling. We never ran Mean, Max or `[CLS]` on GLUE, so we cannot fill their five method rows; those poolers appear only in the noise diagnostic (Appendix I), where they lose by 14–35 points.

Two things are visible here that the delta tables hide. The whole spread on MRPC is 0.37 F1 across eleven arms, and on RTE the seed standard deviation (1.5–3.4) is larger than every gap in the column — which is why we treat both as null. And the bolded best cell is a *different arm* in five of the six columns, which is what a table of noise looks like.

Paired statistics on STS-B: B $\bar{d} = -1.065$, $t = -17.00$, sign $p = 6\times10^{-5}$; BC -1.242, $t = -8.30$, $p = 6 \times 10^{-5}$; ABC -0.837, $t = -6.36$, $p = 0.00098$; AB -0.747, $t = -5.66$, $p = 6 \times 10^{-5}$. All four survive Bonferroni. On CoLA, C gives $\bar{d} = +1.416$, $t = 4.11$, sign $p = 0.0074$ — significant on both tests but above $\alpha = 0.0063$.

Table 3: Matched-estimator comparison: best score at GLOT's own seed (42) over GLOT's own grid, 78 configurations per task, so both columns are computed the same way.

Task	ours	published	Δ
CoLA (MCC)	**50.52**	47.49	+3.03
RTE (acc.)	**59.57**	59.21	+0.36
STS-B (Spearman)	82.99	83.86	−0.87

Table 4: Absolute confirmed scores, mean ± sd over the 15 shared seeds. Metrics follow Mantri et al. (2026): MCC for CoLA, Spearman for STS-B, F1 for MRPC, accuracy for RTE. "—" means the arm was not run on that backbone; the RoBERTa campaign searched a different arm set (published τ and the quantile repair) because its failure mode is thresholding, not geometry. Every number is a confirmation mean, never a tuning maximum.

Arm	BERT-base				RoBERTa-base	
	CoLA (MCC)	STS-B (Spear.)	MRPC (F1)	RTE (acc.)	CoLA (MCC)	STS-B (Spear.)
GLOT (baseline)	45.37 ±1.30	82.53 ±0.33	82.14 ±0.44	53.79 ±3.36	**54.60** ±1.52	83.89 ±0.30
no graph	46.20 ±2.54	82.57 ±0.34	82.17 ±0.31	53.86 ±2.74	54.32 ±2.06	83.84 ±0.32
published τ	—	—	—	—	54.46 ±1.18	**84.21** ±0.29
quantile τ	—	—	—	—	53.19 ±1.59	83.73 ±0.29
A	46.07 ±0.99	82.56 ±0.31	**82.18** ±0.29	53.82 ±1.58	54.37 ±0.89	83.80 ±0.30
B	44.11 ±1.48	81.47 ±0.36	81.81 ±0.39	54.78 ±1.58	—	—
C	**46.78** ±0.79	82.73 ±0.31	81.83 ±0.39	54.34 ±2.07	—	—
AB	44.13 ±1.61	81.79 ±0.46	81.86 ±0.41	54.59 ±2.93	—	—
AC	45.82 ±1.94	**82.77** ±0.20	81.99 ±0.35	55.26 ±1.80	—	—
BC	45.59 ±1.19	81.29 ±0.32	81.83 ±0.43	**55.64** ±1.88	—	—
ABC	44.84 ±1.87	81.70 ±0.34	81.88 ±0.44	54.85 ±1.52	—	—

The graph-free control. We also ran an arm that deletes every edge ($\tau = 0.999$) while keeping the projection, the readout and the same 40 tuning trials. It is statistically indistinguishable from the flat baseline on all four tasks (CoLA $4/11$, $p = 0.12$; STS-B $8/7$, $p = 1.00$; MRPC $6/9$, $p = 0.61$; RTE $7/6$, $p = 1.00$), and on CoLA it is numerically second of eight arms at 46.20, *above* the baseline's 45.37. At this budget the edges contribute little to GLUE accuracy, so any claim about the graph's *geometry* has to clear this bar first. Stage C does not on CoLA (+0.58, $8/7$, $p = 1.00$), though it does on STS-B (+0.16, $12/3$, $p = 0.035$); this is the second reason we do not present C as established. Stage B is significantly *worse* than deleting the graph on CoLA (−2.10, $3/12$, $p = 0.035$), which is the sharpest available statement of its harm.

This does not contradict GLOT's own motivating result. The control here still trains a pooler; it is not a bag of tokens. Against genuine bag-of-tokens baselines the graph earns its place decisively (Appendix I). What the control shows is narrower: on these three GLUE tasks, at 40 trials per arm, the *edges* are not where the accuracy comes from.

F.1 Decomposing Stage C

Stage C's selected configuration differed from the baseline's in two ways at once: a $3.6\times$ denser graph ($q = 0.38$ against 0.10) and a smaller network (2 layers of width 128 against 4 of 256). We therefore ran the 2×2 that crosses *message-passing geometry* with *graph-and-architecture configuration*, on the same 15 seeds. Two cells already existed; the two off-diagonal cells are new runs, written to a separate results file so the released campaign logs are untouched. The hyperbolic cells use C's own tuned curvature settings ($c = 4.0$, HGAT, input clip 0.7), which is the choice most favourable to C.

Each component differences two independently trained cells, so it inherits both their variances; that is why the parts are noisier than the whole. At the observed effect and spread, detecting the

Table 5: Per-stage verdict. Entries are paired mean differences against GLOT's flat baseline on shared seeds. The last column is *not* a fifth task: it is STS-B again on a decoder backbone, so it sits beside the BERT STS-B column and the two read as a replication. † significant on both the paired t-test and the exact sign test; ‡ additionally surviving Bonferroni. *No RTE entry is significant* — that task has 277 evaluation examples and a minimum detectable effect of 2.7 accuracy points, larger than any effect in this paper. "—" was not run.

	BERT-base (encoder)				TinyLlama	
Arm	CoLA (MCC)	STS-B (Spear.)	MRPC (F1)	RTE (acc.)	STS-B (Spear.)	Verdict
A	$+0.70$	$+0.03$	$+0.04$	$+0.02$	$+0.04$	flat everywhere
B	-1.26	$-1.07^{‡}$	-0.33	$+0.99$	—	harmful
C	$+\mathbf{1.42}^{†}$	$+0.20$	-0.32	$+0.55$	—	CoLA only
AB	-1.24	$-0.75^{‡}$	-0.28	$+0.79$	$-3.56^{‡}$	harmful
AC	$+0.45$	$+\mathbf{0.24}$	-0.15	$+1.47$	$+0.21$	flat / slight $+$
BC	$+0.22$	$-1.24^{‡}$	-0.31	$+1.85$	$-1.17^{‡}$	harmful
ABC	-0.53	$-0.84^{‡}$	-0.26	$+1.06$	$-4.51^{‡}$	harmful

Every arm with B is negative on CoLA, STS-B and MRPC. All four are positive on RTE, none significantly.
Every arm with C but not B is positive on CoLA and STS-B.
The two STS-B columns agree in sign on all five arms run on both backbones.

Table 6: BERT-base, wide search, $n = 15$ paired seeds. "p/n" counts seeds positive/negative against the baseline.

	CoLA (MCC)		STS-B (Spearman)	
Arm	Δ	p/n	Δ	p/n
C	$+\mathbf{1.42}$	13/2	$+0.20$	12/3
AC	$+0.45$	10/5	$+\mathbf{0.24}$	11/4
A	$+0.70$	11/4	$+0.03$	9/5
BC	$+0.22$	7/8	-1.24	0/15
ABC	-0.53	6/9	-0.84	1/14
AB	-1.24	4/11	-0.75	1/14
B	-1.26	5/10	-1.07	0/15

interaction at 80% power would take about 65 seeds. This is the cleanest illustration we have of a general hazard: a significant end-to-end improvement can be genuinely irreducible at the sample size that established it, and reporting any one of these components as "the" explanation would have been unsupported.

G Porting to other backbones

The failure is layer-dependent: ModernBERT reaches 0.599 at the final layer but 0.996 at layer 12. Mantri et al. (2026) do not state which layer they pool, so "which backbones are affected" cannot be settled from the published description; we treat RoBERTa as the one clear case.

Forcing the graph sparse costs 1.41 MCC ($t = -3.44$, $2/13$, surviving Bonferroni at $\alpha = 0.0125$). Against `paper_tau` directly the repair loses on both tasks (-1.27 CoLA, -0.48 STS-B, the latter $2/13$, $p = 0.0074$).

Density and scale are separate. We had credited a $21.71 \rightarrow 27.15$ MCC recovery on ModernBERT/CoLA to density-matching plus median norm rescaling, measured at one seed with both factors moved together. The full 2×3 factorial over 5 seeds (Table 11) separates them. The choice of rescaling statistic matters far more than whether you rescale, and it is the only effect here significant on both tests: on ModernBERT layer 12, whose mean/median norm ratio is 7.66, mean-based

Table 7: Density × curvature factorial, CoLA MCC, mean ± sd over the 15 shared seeds. Realised edge density was verified per cell: 0.0999 in the base-config row, 0.3759 in the C-config row, so the columns really are density-matched.

	Euclidean MP	hyperbolic MP
base config (q=0.10, K=4, h=256)	45.37 ±1.30	45.77 ±1.42
C config (q=0.38, K=2, h=128)	45.36 ±2.19	**46.78** ±0.79

Table 8: Exact decomposition of Stage C's headline gain, paired over 15 seeds. The three components sum to the total. The total is significant on both tests; none of its parts is.

Component	Δ	sd	t	p/n	sign p
geometry alone (at base config)	+0.400	1.75	0.88	8/7	1.0000
configuration alone (Euclidean MP)	−0.012	2.55	−0.02	10/5	0.3018
interaction	+1.028	2.95	1.35	10/5	0.3018
total (C − baseline)	**+1.416**	1.33	**4.11**	**13/2**	**0.0074**

(`rms`) rescaling costs 11.03 MCC ($t = -5.47$, $0/5$) — the mean is exactly the statistic outliers corrupt. Median rescaling is directionally positive (+5.96 and +5.32 on layer 12, +1.34 at the final layer with $5/0$) but reaches significance nowhere at $n = 5$. Density-matching has no consistent sign (+4.58, −1.91, −0.25, +0.45). BERT is inert throughout, every contrast at most 0.27 MCC — our earlier single-seed reading of that control (45.54 → 43.41) was noise.

H DECODER BACKBONE

TinyLlama-1.1B on STS-B, $n = 15$ shared seeds, chosen over SmolLM2 on measured geometry (0.073 density against BERT's 0.103, whereas SmolLM2 sits at 0.741). Minimum detectable effect 0.503.

Arm	mean	std	Δ	sign p	pos/neg
AC	80.15	0.54	+0.205	0.302	10/5
A	79.99	0.51	+0.043	0.607	9/6
baseline	79.95	0.49	*ref*	—	—
BC	78.78	0.95	−1.165	0.00098	1/14
AB	76.39	1.87	−3.556	6×10^{-5}	0/15
ABC	75.44	3.58	−4.506	6×10^{-5}	0/15

The ordering matches BERT, and the standard deviation grows in step with the number of hyperbolic parts (0.48 → 0.95 → 1.87 → 3.58).

I NOISE DIAGNOSTIC

We reproduce the signal-in-noise test of Mantri et al. (2026): a short signal phrase carrying a logical dependency, injected into random distractor words, with the distractor ratio swept from 20% to 90%. BERT, mean ± std over 5 seeds; the original reports one seed.

Table 9: Edge density from the published grid $\tau \in \{0.1, 0.3, 0.6\}$, with the tenth-percentile token cosine that produces it. Measured on 192 CoLA sentences at the final layer, padding masked and self-loops excluded. This is a property of the backbone and τ alone. Density near 1 means the GNN sees a nearly complete graph.

Backbone	cos p10	τ=0.1	τ=0.3	τ=0.6
TinyLlama-1.1B	0.016	0.762	0.533	**0.073**
ModernBERT-base	0.076	0.866	0.759	0.599
BERT-base	0.098	0.876	0.505	**0.103**
SmolLM2-360M	0.322	0.956	0.908	0.741
RoBERTa-base	**0.693**	1.000	1.000	**0.991**

Table 10: RoBERTa-base, $n = 15$ paired seeds, tuned per arm and per task on RoBERTa itself (40 trials each). `paper_tau` pins the published grid and gives the near-complete graph above; `density_fix` is the quantile repair. The repair is worst on both tasks, and on CoLA its deficit survives Bonferroni — despite having *won* the tuning stage (56.77, best of five).

	CoLA (MCC)			STS-B (Spearman)		
Arm	mean	Δ	p/n	mean	Δ	p/n
baseline	**54.60**	*ref*	—	83.89	*ref*	—
`paper_tau`	54.46	-0.14	8/7	**84.21**	$+\mathbf{0.32}$	12/3
A	54.37	-0.23	5/10	83.80	-0.09	5/10
`density_fix`	53.19	$-\mathbf{1.41}$	2/13	83.73	-0.15	5/10

Arm	20%	50%	80%	90%
AC	95.0 $\pm$3.4	88.2 $\pm$11.2	90.8 $\pm$5.8	**88.5** $\pm$8.9
C	95.0 $\pm$3.3	87.6 $\pm$11.2	90.8 $\pm$5.9	88.4 $\pm$8.8
A	**97.0** $\pm$1.3	**91.9** $\pm$4.1	**91.0** $\pm$2.7	84.9 $\pm$9.1
baseline	96.0 $\pm$0.5	90.2 $\pm$5.0	88.1 $\pm$4.7	79.8 $\pm$10.8
B	95.4 $\pm$1.4	88.7 $\pm$2.8	83.8 $\pm$13.8	75.5 $\pm$14.9
Mean	73.5	63.2	64.0	57.4
`[CLS]`	70.0	61.4	62.1	65.4
Max	63.0	55.2	53.1	51.8
Mantri et al. (2026) Table 7, BERT row (single seed):				
GLOT	97.2	97.0	97.8	98.8

GLOT's advantage over bag-of-tokens pooling is real and holds in every seed: mean, `[CLS]` and max lose 14–35 points at every ratio, $0/5$ seeds positive. Stage B is again last among the hyperbolic arms.

We do not reproduce the published trained-pooler numbers. At 90% distractors the original reports 98.8 where we measure 79.8 ± 10.8 for the same method, and their curve is flat or rising under noise where ours falls. The divergence is not uniform: the static poolers agree closely (-2.2 to $+5.6$ across twelve cells), while both *trained* poolers diverge in opposite directions, so the 37-point margin their Table 7 reports for GLOT over AdaPool at 90% becomes -3.9 in our hands. Their Appendix B.4 does not give the vocabulary source, the signal templates, or the pooler budget, so we cannot localise the cause. We flag this as unresolved, and we never use the published values as a baseline.

J GNN CHOICE AND TRAINING RECIPE

GNN backbone. Over 5 seeds at the published recipe, CoLA gives GAT 45.38 ± 1.19, GCN 45.14 ± 1.78, GIN 44.86 ± 1.33; STS-B gives 82.69 ± 0.34, 82.46 ± 0.27, 82.47 ± 0.32. Spread

Table 11: Density $\times$ scale factorial, CoLA MCC, mean $\pm$ sd over 5 seeds. Columns vary the threshold, rows vary token-norm rescaling. At $n = 5$ the exact sign test cannot fall below $p = 0.0625$, so nothing here can meet our both-tests bar; we report it as a lead.

Backbone (layer)	Rescaling	$\tau = 0.6$ (pub.)	$q = 0.05$
BERT-base (final) *control*	none	45.25 ±0.97	45.00 ±1.51
	`rms`	45.01 ±1.25	45.03 ±1.18
	`median`	44.98 ±1.00	45.08 ±0.83
ModernBERT-base (L12)	none	18.64 ±10.16	23.22 ±6.17
	`rms`	7.61 ±7.71	19.18 ±2.98
	`median`	24.60 ±7.12	**28.54** ±2.41
ModernBERT-base (final)	none	37.53 ±1.09	35.62 ±6.00
	`rms`	38.64 ±1.48	37.61 ±1.72
	`median`	**38.87** ±1.18	37.57 ±1.79
RoBERTa-base (final)	none	52.39 ±1.98	52.89 ±0.97
	`rms`	52.45 ±1.43	–
	`median`	**52.54** ±1.38	–

0.52 MCC and 0.23 Spearman, inside seed noise, and GAT is the only one of the three that consumes edge attributes at all.

The STS-B residual is not a recipe mismatch. The released README trains differently from the paper's Appendix B.2 (3 epochs, `scorer_hidden` 256, $\tau = 0.8$ against 2, 128, 0.6). We ran the $2 \times 2 \times 2$ decomposition at seed 42. Epoch count is the only factor that moves anything ($82.36 \rightarrow 82.66$); τ and `scorer_hidden` are inert to within 0.04. Over 5 seeds the two recipes are indistinguishable (82.69 ± 0.34 against 82.69 ± 0.23). The best cell, 82.66, is still 1.20 below the published 83.86, so the residual stands.

Layer choice overfits. Layer 8 beat layer 12 by +4.62 MCC on CoLA but only +0.25 on STS-B, and lost on RTE (−0.36) and MRPC (−0.46).

K A RETRACTED RESULT OF OUR OWN

Under the graph-only search, Stage A on STS-B gave $\bar{d} = +0.223$, $t(14) = 6.20$, sign $p = 6 \times 10^{-5}$, 15/15 seeds positive, and replicated on 12 held-out seeds. It does not survive the wide search: $\bar{d} = +0.028$, $t = 0.69$, 9/5. The effect was real but its cause was not geometry — the baseline had been pinned at $K = 2$, $h = 128$ while nothing else was, and it recovers the whole gap once it can choose $K = 4$. This is the clearest illustration of the paper's own point: a pooling comparison is only as good as the budget given to its control.

L LONG-TEXT AND RETRIEVAL BENCHMARKS

Mantri et al. (2026) also report IMDB (long-text classification) and seven MTEB tasks (zero-shot retrieval and clustering). We ran both, early in the project, at a single seed and on BERT only. We report them here for coverage and draw no statistical conclusion: with $n = 1$ there is no test to run.

One of these rows is not what its name says. These runs predate our switch to quantile thresholding. The `A_threshold` arm used an absolute radius $\rho = 1.0$ on unnormalised BERT features, and as Appendix M explains, real Poincaré distances there lie in $[8.85, 11.76]$ — so that arm trained on an *empty graph*. We therefore relabel it: it is not Stage A, it is an accidental graph-free control. Read that way it replicates, on two further benchmark families, the GLUE finding of Appendix F that deleting every edge costs nothing: +0.24 accuracy on IMDB and −0.06 on the MTEB mean. `A_knn` used a k-NN Poincaré graph and is a genuine Stage A variant; it is also flat.

Table 12: IMDB accuracy and MTEB main scores ($\times$100), BERT-base, seed 42, one run per cell. "no graph†" is the mislabelled `A_threshold` arm described above. MTEB is zero-shot after contrastive training on MS MARCO triplets; the seven tasks are those of Mantri et al. (2026).

	GLOT (cosine)	no graph† (empty)	Stage A (k-NN)
IMDB (acc.)	92.26	**92.50**	92.46
Banking77Classification	46.91	46.90	**46.91**
STS12	30.02	**30.04**	27.68
STS13	59.57	59.60	**59.83**
SciFact	13.93	**14.10**	13.39
ArguAna	**28.25**	28.11	27.73
TwentyNewsgroupsClustering	24.49	24.13	**25.10**
SprintDuplicateQuestions	**34.67**	34.57	34.40
MTEB mean	**33.98**	33.92	33.58

Across seven MTEB tasks the largest gap to the baseline is 0.17 for the graph-free arm and 2.34 for k-NN (both on tasks where all three arms sit within a few points of each other). Nothing here is separable at one seed, and neither Stage B nor Stage C was ever run on these benchmarks. Doing it properly is expensive: our IMDB runs took 3.8 hours each at 512 tokens, so eight arms $\times$ 15 seeds would be roughly 20 days.

M Notes for reproducers

Three details of the released code cost us time. None affects the conclusions of the original paper.

- `jk_mode=lstm` is accepted by the parser but crashes at runtime, and the advertised `mean` and `none` are not implemented. The usable space is {`cat`, `max`}.
- Thresholding Poincaré *distance* at a fixed radius gives an empty graph on real features (see §2). The signature is a score that is bit-identical across a whole threshold grid. We use a quantile threshold everywhere for this reason.
- The logged edge-density counts self-loops in the numerator but excludes the diagonal from the denominator, so an *empty* graph logs $1/(n-1) \approx 0.10$ at CoLA's typical length rather than 0. This is a reporting defect only — no training path reads the statistic — but it is a trap for exactly the audit this paper performs. It was corrected partway through the project, so the CoLA logs carry the offset and the later MRPC logs do not; densities are not comparable across campaigns without checking. Any density we quote is re-measured with the standalone instrument of Table 9, never taken from training telemetry.

N Structural graph variants

Since the geometric arms are flat on CoLA, MRPC and RTE, we also tried a *structural* prior: positional edges joining tokens with $|i-j| \leq w$, positional-only graphs, their union with Stage A, and k-NN graphs. Tuning maxima over 10 trials: baseline 45.32, POS 47.08, POS_ONLY 46.80, A_POS 44.87, KNN 45.32.

These are tuning maxima and are not results. At CoLA's noise level ($\sigma = 2.09$) the expected inflation of a best-of-10 is ≈ 3.22 MCC, which exceeds every gap in the list including POS's apparent $+1.76$ lead. We report them because the arms are implemented and released, but draw no conclusion. Confirming them on 15 paired seeds is the most obvious extension we did not run.

O LIMITATIONS

- **Stage C's gain is real but unattributed.** The 2×2 of Table 7 decomposes $+1.42$ MCC exactly into geometry ($+0.40$), configuration (-0.01) and interaction ($+1.03$), and none of the three is significant at $n = 15$ even though the total is. Resolving the interaction needs about 65 seeds. Until then we can say the effect requires both the denser graph and the curved message passing, but not which is responsible. It also does not reach MRPC, does not separate from the graph-free control, and was never run on the decoder.
- **We cannot say why the edges contribute so little on GLUE.** The graph-free arm matches the baseline on all three tasks, yet the graph is decisive on the noise diagnostic. Reconciling those two observations — most likely by identifying which inputs actually need relational pooling — would say more about GLOT than anything in this paper. We did not attempt it.
- **We select the epoch on the split we report**, because GLUE test labels are not public. This is a smaller version of the practice we criticise in §3. Splitting the development set into selection and reporting halves would remove the objection, and is the single most valuable improvement to this protocol.
- **One reproduction residual is unexplained.** STS-B stays 0.87 short under every estimator we tried; we tested and ruled out the leading hypothesis (Appendix J).
- **The scale factorial is underpowered.** At $n = 5$ the exact sign test cannot fall below $p = 0.0625$, so no contrast in Table 11 can meet our own bar.
- **Non-Euclidean layers may need structure-aware optimisation.** We gave every arm the full grid, and the search did move the learning rate per arm and per task rather than pinning it to a flat-tuned value. But we did not implement per-manifold Riemannian optimisation or curvature-aware initialisation. This is the strongest available objection to our Stage B negative, and we state it rather than merely acknowledging it.
- **Every task we ran is small, and that may explain Stage B.** CoLA, STS-B, MRPC and RTE have 8,551, 5,749, 3,668 and 2,490 training examples. Stage B hurts by amplifying seed variance, which more data suppresses, so its harm may not survive at scale — and RTE, the smallest, is the one task where B-containing arms trend positive. We could not test this: at 495 runs per task our design costs an estimated 170 hours for SST-2, 264 for QNLI, and 38 and 41 days for QQP and MNLI. A reduced design — four arms, confirmation seeds only, reusing the CoLA-selected configuration — brings SST-2 to roughly 21 GPU-hours and is the single most valuable experiment this paper is missing.
- **RTE is underpowered and contributes nothing.** With 277 evaluation examples and a seed standard deviation of 3.36, its minimum detectable effect is 2.7 accuracy points. All seven hyperbolic arms are nominally positive there and not one is significant. We include it for completeness and base no conclusion on it.
- **Coverage.** Mantri et al. (2026) report ten GLUE tasks on six backbones, plus IMDB and seven MTEB tasks. We report four GLUE tasks with the full nine-arm design on BERT, two on RoBERTa, one on TinyLlama, a synthetic noise task, and IMDB plus the same seven MTEB tasks at a single seed for the graph stage only (Appendix L). We never ran Mean, Max or `[CLS]` on GLUE, so we cannot reproduce their method-comparison table; our rows are GLOT's own components.